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ROLLS ROYCE	**ENGINEERING SPECIFICATION**	**RRES 90008**
		Revision B
		First Issued 19 June 12
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Balancing of Gas Turbine Rotors and Components

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1 SCOPE:

1.1 This Engineering Specification documents the methods required for the application of a balancing process.

1.2 Where a procedural conflict occurs with older drawings and manuals, this document takes precedence as it contains all the latest practice that is considered necessary for a sound balancing procedure. Note that changes to the engineering intent (limits etc.) will require engineering approval via a suitable concession process.

1.3 This document assumes a basic knowledge of balancing theory and practice. Further information and training material is available on the Rolls-Royce Balancing Website (https://rollsroyce.sharepoint.com/sites/engineroom/en-gb/balancing/Pages/default.aspx). Where access to the Rolls-Royce Balancing Website is not possible, the referenced calculators and other additional material are available on request.

1.4 Procedures that do not conform to this specification will require buy-off from a specialist balancing engineer.

1.5 For the purposes of this document all units are SI standard.

1.6 The procedures within this document are designed for the Hard Bearing balancing machine type, unless otherwise stated.

1.7 Topics that are not covered by this document:

a) Detailed information on operating balancing machines.
b) Detailed information on verification testing of balancing machines.
c) Balancing theory that is covered by a training course.

2 REASON FOR USE:

2.1 To identify the requirements necessary for the control of a balancing procedure. This is to ensure that any rotor to be balanced is subjected to the correct procedures in order to meet the design intent as specified by component definition.

3 APPLICABLE DOCUMENTS:

The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

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3.1 ISO publications:

Available from the International Organization for Standardization (ISO) 1, ch. de la Voie-Creuse, Case postale 56 CH-1211 Geneva 20, Switzerland **www.iso.org** and internally through Engine room A to Z links:

ISO 1925	Mechanical vibration balancing vocabulary. (superseded by ISO 21940-2)
ISO 1940-1	Mechanical vibration quality requirements for rotors in a constant (rigid) state. (superseded by ISO 21940-11)
ISO 2953	Balancing machines, description and evaluation (superseded by ISO 21940-21).
ISO 7870-2	Shewhart Control Charts
ISO 8258	Shewhart Control Charts (superseded by ISO 7870-2)
ISO 11342	Mechanical vibration methods and criteria for balancing flexible rotors. (superseded by ISO 21940-12)
ISO 21940-2	Mechanical vibration, rotor balancing, part 2, vocabulary.
ISO 21940-11	Mechanical vibration, procedures and tolerances for rotors with rigid behaviour.
ISO21940-12	Mechanical vibration, procedures and tolerances for rotors with flexible behaviour.
ISO 21940-13	Criteria and safeguards for the in-situ balancing of medium and large rotors.
ISO 21940-14	Procedures for assessing balancing errors
ISO 21940-21	Description and evaluation of balancing machines.
ISO 21940-23	Enclosures and other protective measures for the measuring station of balancing machines.
ISO 21940-31	Susceptibility and sensitivity of machines to unbalance.
ISO 21940-32	Shaft and fitment key convention

3.2 Society of Automotive Engineers (SAE) Aerospace Recommended Practices (ARP) Aerospace Standards (AS) and Aerospace Material Specifications (AMS) and internally through Engine room A to Z links:

Available from SAE, 400 Commonwealth Dr., Warrendale, PA 15096, USA:

SAE ARP4048	Description and evaluation of 2 plane horizontal machines. (superseded by SAE AS 8617)
SAE ARP4050	Description and evaluation of 2 plane vertical machines. (superseded by SAE AS 8617)
SAE ARP4162	Balancing machine Proving Rotors.
SAE ARP5323	Description and evaluation of 1 plane vertical machines. (superseded by SAE AS 8617)
SAE AS8617	Balancing Machines – Verification Test Requirements.

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4 DEFINITION OF TERMS:

3D Moment Weighing or Weight:
When weighing a blade and determining the Centre of Gravity in three dimensions: Radial, Tangential and Axial.

Arbor:
A tool or mandrel used for locating the rotor on a balancing machine. Arbors have a running diameter that is true to the axis of rotation. They are not intended to simulate rotor properties.

Artifact Rotor:
An engine part or representative rotor which is used to confirm balancing machine performance.

Balancing:
A procedure by which a mass distribution of a rotor is checked and, if necessary, adjusted to ensure that the residual unbalance/vibration at the bearing planes is maintained within acceptable limits.

Blade Moment Weighing or Weight:
The mass weight of the blade multiplied by its radial CG position.

Couple Unbalance:
Where the unbalance causes an inclination of the principal axis of inertia, which passes through the CG.

Critical Dimension:
With respect to balancing Simulators, a physical dimension that affects its function

Dynamic Unbalance:
Where the unbalance causes an inclination of the principal axis of inertia, which doesn't pass through the CG and is caused by a combination of static and couple unbalance.

Flexible Rotor:
Where the unbalance state changes within the operating speed range due to elastic/plastic deformation.

Grout:
A bonding medium that fills the gap between the bed of the machine and its inertia block.

Hard Bearing Balancing Machine:
Where the dynamometer of the machine has sufficient stiffness such that it reacts to force. The unbalance is measured below its critical speed. This type of machine is more desirable than the soft bearing machine as it is permanently calibrated and has a minimal set up time. The hard bearing machine is the preferred standard for balancing gas turbine parts and assemblies.

Initial Unbalance:
The unbalance that exists in the rotor prior to balancing.

Mass Weighing or Weight:
The vertical force exerted by a mass as a result of gravity in grams.

Master Calibration Blade:
A blade (compressor or turbine) which is used during a moment weighing process. The blade is used to certify Working Calibration Blades.

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May:
Denotes a permissible practice or action.

Must:
Is used only for legislative or regulatory requirements (e.g., Health and Safety) and shall be complied with.

Proving Rotor:
A solid rotor that is designed specifically for testing balancing machinery. The rotor has accurately positioned test planes where known weights are applied.

Residual Unbalance:
The unbalance that exists after correction.

Repeatability
The unbalance change that occurs when a part is removed and re-fitted to its tooling or mating part.

Rigid Rotor:
A rotor that can be balanced without any consideration of deformation within its operating speed range.

Rotor Specific Calibration:
A calibration process, which bypasses the permanent calibration of the machine and is specific to the job being balanced.

Scatter
The unbalance variation that occurs in between balancing runs/measurements.

Shall:
Denotes a mandatory requirement. Departure from such a requirement is not permissible without formal agreement with the project or SCU technical authority (e.g., Chief Engineer, SCU Chief Design Engineer) and shall be properly documented.

Should
Denotes a recommendation or advice and is expected to be followed unless good reasons are stated for not doing so.

Simulator:
A rotor designed for balancing purposes only, which replicates the mass, CG & inertia properties of an engine part. Simulators are also known as Dummy Mandrels and PMIs (Polar Moment of Inertia).

Soft Bearing Balancing Machine:
Where the dynamometer of the machine has sufficient flexibility such that it reacts to displacement. The unbalance is measured above its critical speed. This type of machine requires calibrating before use.

Static Unbalance:
Where the unbalance causes a parallel displacement of the principal axis of inertia.

Swash:
Where the end-face/surface of a part is not perpendicular to its rotational axis.

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Trim Balancing:
A method of balancing in-situ e.g. During engine test or during engine installation in the aircraft.

Unbalance:
A condition that exists in a rotor when vibration is generated by centrifugal force. It is convention that the angle of the unbalance defines the heavy position of the vibration Source.

Working Calibration Blade:
A blade that has been calibrated against a Master Calibration blade and is used for setting up a moment weighing system.

5 ABBREVIATIONS AND ACRONYMS:

CG - Centre of Gravity
e_{mar} - declared accuracy of the balancing machine in gmm/Kg.
Id - Diametral Moment of Inertia
Ip - Polar Moment of Inertia
NLGI - National Lubricating Grease Institute
SPC - Statistical Process Control
UCL - Upper Control Limit

6 BALANCING MACHINE - GENERAL PRACTICES:

6.1 Balancing machine selection:

6.1.1 During the planning stage of a balancing process, it is imperative to consider the machinery required to measure and correct the unbalance to the required standard. The following points apply:

a) The type of balancing machine; hard or soft bearing. Although the hard bearing machine is the preferred standard, there are older soft bearing machines still in use. Also, there are cases where tooling has been designed for use in a soft bearing machine a later adapted for use in a hard bearing machine. This can cause measurement problems as some designs employ deep groove ball bearings, which can generate rotational signals when not aligned correctly. Also note that some older soft bearing systems employ Tie Frames (used to link the bearing journals) and they may not perform correctly in a hard bearing balancing machine.
b) The type of balancing machine; rotating or non-rotating. Where it is not necessary to rotate a part about a surface, satisfactory balancing can be achieved using a non-rotating machine. This offers some significant advantages over the rotating type and it can also be used for moment weighing requirements.
c) The job to be balanced shall be within the specification range of the machine as specified by the manufacturer.
d) The bearing journal diameters of the job should be within the range of the scale on the machine's Roller Bridge. If this is not available, the diameter of the machine's Rollers shall not be within 10% of the bearing journal diameter of the job.
e) Where a job is to be shaft-driven, the rotor assembly should be inline and level with the machine drive. A small inclination of the drive is acceptable, providing it has no effect on the unbalance measurements
f) The measuring system should have sufficient functionality for the planned job e.g., indexing & averaging etc.
g) The machine (hard bearing type) shall be tested for calibration. See the section on machine verification.
h) The balancing machine drive system should adequately control RPM and acceleration.

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6.1.2 When procuring new balancing machinery, the machine requirements should be clearly defined within a specification document and requirements register. The document shall include a clear definition of the machine's functionality/requirements. This information shall be impartial and encourage custom solutions where appropriate. An example of this is provided on the balancing website.

6.2 Balancing machine installation:

6.2.1 To obtain the best performance from a balancing machine (hard type) it is imperative that it is correctly attached to a sufficiently sized mass/inertia block. This is essential to the dynamometer of the machine, which requires an effective ground in order to react correctly. Inertia blocks shall be firmly supported without any form of expansion joint, which could permit movement of the block. Some smaller machines are designed with their own inertia mass and do not require a specific installation.

6.2.2 Alternatively, balancing machines can be installed directly to a floor slab although this is less desirable to a purposely designed inertia block. In either case the slab or base shall have known mass and density as required by the manufacturer/supplier of the machine. Also note that there are minimum dimension requirements to site machines away from the edge of a floor slab.

6.2.2.1 Although balancing machines will generally ignore non-synchronous vibration it is advisable to select a site, which is not affected by high levels of vibration.

6.2.2.2 During installation, particular attention is required to ensure that the machine is the correct level. The gap underneath the bed shall be correctly grouted such that when fully cured there are no cracks or structural weakness. If the machine is installed correctly the static load of the machine will be evenly distributed across the grout surface. It is therefore important to follow the manufacturers' requirements precisely and if necessary, seek specialist advice. Any machine that has abnormal performance issues and visible defects (e.g. cracking) with the bed grout should be removed and re-installed correctly.

6.2.2.3 With new installations it will be necessary to conform to all the relevant health and safety requirements, specifically noise generated by the rotor or engine part being balanced. Where a machine is to be positioned inside an acoustic cabinet, the cabinet shall not be secured to the bed of the machine. Acoustic cabinets shall not influence the measurement of unbalance in any way and shall not inhibit inspection of the machine bed.

6.3 Balancing machine verification test requirements:

6.3.1 All balancing machines shall be tested with a Proving Rotor at intervals not exceeding 12 months. On horizontal machines, it is important to test the machine with the rollers that are the most frequently used. The size and type of Proving Rotor used for the verification shall be chosen with consideration of the jobs to be balanced. The mass of the Proving Rotor should be approximately representative of the average mass of the jobs being balanced. Where there is a large range in job mass, the machine should be tested with two Proving Rotors with a mass biased to the lower and top one third of the machine's range capacity (Ref. ISO 21940-21). The test weights used for the verification test shall have a maximum e_{mar} value of 1gmm/Kg.

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The verification shall comply with either of the following standards:

Notes:

1. It is not necessary to carry out the Reference Change and the Compensator test.
2. The balancing results/data should be assessed using a calculation-based process that calculates unbalance change.
3. When calculating unbalance change for the URR test, the values generated by the travelling weight shall be within 5% of the predicted values.
4. Where a machine (horizontal type) has alternative methods of drive (shaft/belt) the test shall be carried out on both drive systems.
5. There are spreadsheet calculators available on the Balancing Website.

ISO 2953	Balancing machines, description, and evaluation (superseded by ISO 21940-21)
ISO 21940-21	Description and evaluation of balancing machines.
SAE AS8617	Balancing Machines – Verification Test Requirements

6.3.2 An artifact rotor may be used for checking balancing machine providing it has test planes and weights which conform to ISO/SAE standards. Where there is a requirement to compare one machine against another, the rotor should be seeded with static unbalance; equivalent to five to ten gmm/kg of rotor mass. The residual static unbalance (gmm) should be repeatable within a value equivalent to1 gmm/Kg of rotor mass.

6.3.3 Moment Weighing machines shall be verification tested in accordance with Appendix A at intervals not exceeding 12 months.

6.4 Balancing machine maintenance:

6.4.1 During day-to-day usage, it is important that specific features of the machinery are monitored for wear and damage:

a) Check the pedestal rollers for wear and uneven running. It is imperative that the rollers are correctly crowned and show a running witness mark in the centre of the contact area with the rotor. The surface of the rollers shall not have any defects that can be felt with a 1 mm stylus
b) Check bearing housing saddles for general damage. The working parts of the saddle shall be clean and free of any contamination. Saddles that utilise ball bearings shall be free to operate by hand without stiction. There shall be no clearance in the bearings when applying light (negative) hand pressure.
c) Drive shaft joints should not have any noticeable wear when applying hand pressure.

Note: These checks are in addition to scheduled checks that are specified by the manufacturer of the machine.

6.5 Selecting machine rollers:

6.5.1 The rollers that are supplied with the balancing machine pedestals are crucial to the performance of the machine. These are supplied by the manufacturer of the balancing machine and are specifically designed for balancing purposes. The crowning on the outer diameter is crucial to the alignment of the rotor. When this is degraded rotational forces are interpreted as unbalance resulting in an incorrectly balanced rotor or part.

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6.5.2 The bearings are sized for different bearing journal diameters and are provided with a scale to aid alignment with the drive of the machine. The roller diameter shall not be within 10% of the bearing journal diameter on the job or be exactly 50% of the diameter on the job. This will avoid any synchronous signals being interpreted as unbalance.

7 TOOL DESIGN:

7.1 General principles:

7.1.1 Tooling that can influence the unbalance measurements shall be designed such that the final unbalance limit of the part or job can be met. There shall be sufficient margin to ensure that the job remains balanced and within drawing limits, regardless of its angular relationship with any tooling. The following design principles apply:

a) Rotating tooling shall be completely rigid and have no effect on the balancing results.
b) Rotating tooling shall be designed such that unbalance errors can be identified and managed. The tooling shall be capable of indexing relative to the job a suitable angle (ideally 180°) so that its errors can be compensated for.
c) Surfaces that abut engine parts shall have a minimum hardness of 45 HRC in order to control tool wear within its design life.
d) Tooling shall be designed such that it does not damage the job to be balanced in any way.
e) The gross mass of rotating tooling (excluding Simulators) should not exceed the mass of the rotor. Where this is not possible, the gross mass should be kept to a minimum.
f) The parasitic mass of static parts should be kept to a minimum.
g) Painted parts shall be of sufficient quality to prevent flaking and contamination of working parts.
h) Plain bearings shall not be used for locating the rotor in hard-bearing balancing machines.
i) Deep-groove ball bearings should not be used for locating the rotor in hard-bearing balancing machines.
j) Roller type bearings can be used in tooling designs providing they do not have excessive clearance resulting in movement of the rotor. Diametrical clearance should be between 0.013 mm and 0.039 mm. Note that bearing assemblies designed for engine use have larger clearances and in general are not suitable for balancing equipment at lower speeds.
k) Self-contained bearings used in balancing equipment shall not have free location bearing elements i.e., the bearing elements shall be located in a cage.
l) Static and rotating tooling for hard bearing balancing machines shall not have a natural frequency within the running range of the machine; ideally the natural frequency should be at least 3 times the balancing speed.
m) Rotating tooling shall not be designed with moveable parts that affect the repeatability. Securing clamps etc. shall be located such that any unbalance change does not affect the ability to meet the final unbalance limit.
n) Bearing location surfaces that are used to locate the rotor on the balancing machine shall meet the following requirements:
   Surface finish – <0.4 microns Ra
   Surface hardness – >45 HRC
   Total run-out across the whole surface – <12 microns
o) Any restraining feature that stores energy e.g., under tensile preload or hydraulic pressure shall be designed such that failure of any part is contained. Where this is not possible, the design shall be subject to a thorough risk analysis in order to satisfy health and safety requirements.

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p) Rotating tooling shall be balanced such that the tooling influence on the job is less than ten times the final per plane balance limit on the job.
q) Rotating tooling used for dynamic balancing shall have two correction planes to correct for couple imbalance.

7.2 End-stop control bungs:

7.2.1 On some jobs that need to be belt-driven, there is not always a suitable surface for setting the end-stop rollers of the machine. This can be overcome with a suitable bung made from Acetal, which is light and has little effect on the job unbalance. See 1. The bung should have suitable tolerances so that it can be fitted easily on an internal or external diameter. Fixing a central ball bearing provides a location point for the end-stop at the axis of rotation. This avoids any undesirable once-per-rev signals.

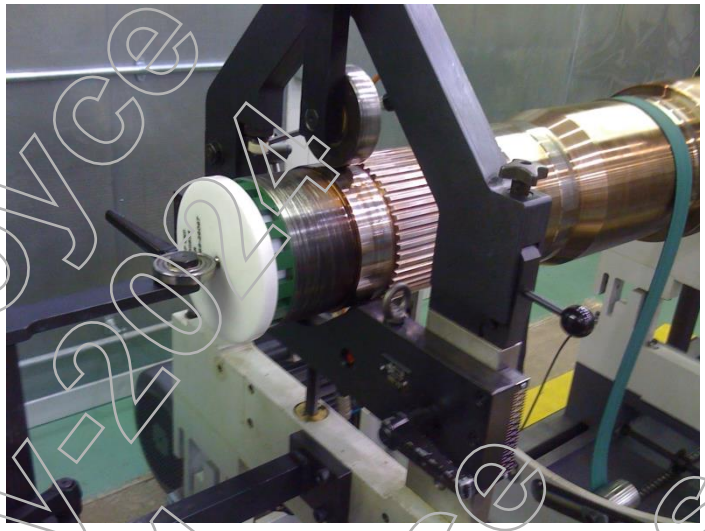

FIGURE 1 - EXAMPLE OF AN END-STOP BUNG MADE FROM ACETAL

7.3 Balancing machine roller bridges – design features:

7.3.1 With some jobs it is not always possible to utilise the open rollers supplied with the machine, which invariably is due to difficulty in accessing the bearing surface. This can be overcome by repositioning the rollers on suitable supports. See Figure 2. The following design principles apply:

a) The minimum tolerance class of the bearings shall conform to ISO 492 class 6.
b) The lubricant used within the bearing should not be packed with grease that has a viscosity greater than 2 (NLGI) as a higher number is likely to cause skidding.
c) The contact surface of the rollers shall be crowned. The outside diameter of the bearing should have a 1500 mm crown radius. On smaller bearing diameters (less than 35 mm approximately) the crown radius may be reduced to 1000 mm. Where bearings have a reduced width (less than 10 mm approximately) the crown radius should be reduced to 500mm to ensure that the rotor contact surface does not contact the edges of the bearing surface. The bearings shall be checked to ensure that there are no resonant frequencies within the operating range of the machine.
d) The bearing supports should be permanently fixed and have sufficient stiffness to prevent any measurable deflection of the job. Failure to meet this requirement is likely to corrupt the unbalance readings necessitating a rotor specific calibration of the job.

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e) Bearing support pillars should have a heavy interference with the bearing internal diameter of 0.001 mm per 1.0 mm of diameter.
f) The fixing centres of the bearings should be designed such that the angular relationship with the rotor axis is equal (i.e., 60°).

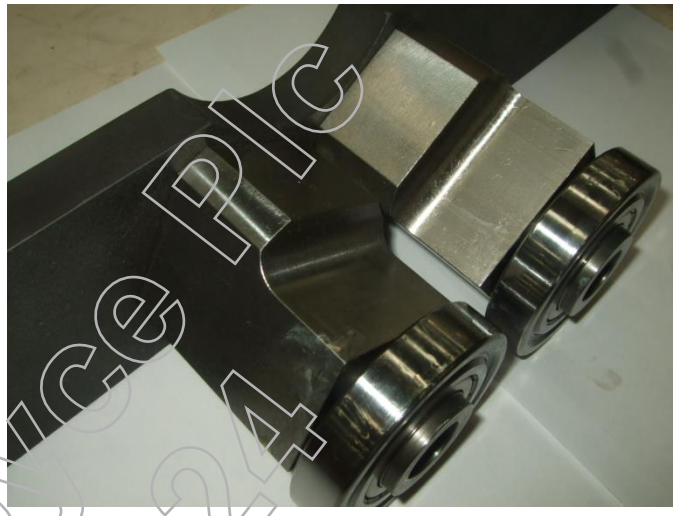

FIGURE 2 - EXAMPLE OF A ROLLER-BRIDGE, DESIGNED TO FIT A SCHENCK HL4 BALANCING MACHINE

7.4 Moment weighing fixtures for blades:

7.4.1 To locate an engine compressor or turbine blade in a Moment Weighing machine will require a custom made fixture. The fixture is required to adapt the blade root feature to the location features of the machine. When designing a fixture for moment weighing, the following principles apply:

a) The blade shall be located such that the correct moment is measured. When measuring the radial moment the working surface of the blade should be in the vertical plane. The blade should be located about the same surfaces as it is in the engine part.
b) Blades with dovetail or pin type roots should be loaded radially outwards to ensure measurements are representative of the engine running condition.
c) Blades with fir-tree type roots may be located on opposing knife-edges or pins providing they contact a working surface of the root.
d) Where an absolute unbalance measurement is required, the fixture shall be designed such that measurements are taken with the blade at its running radius.
e) When removing and re-fitting the blade to the fixture, the measurements should repeat within 0.05%.
f) The contact surfaces of the fixture shall have sufficient hardness to avoid wear-out and variation in readings.
g) Any moving parts of the fixture shall not have any significant effect on the measurements.

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7.5 Simulators – requirements:

7.5.1 Where a design definition requires a Simulator, the required properties will be specified by a drawing or solution. The Simulator should be designed using suitable design/modelling software. The following design principles apply:

7.5.1.1 Properties:

a) Mass shall be within 5% of the design definition requirement.
b) Centre of gravity axial position shall be within 5% of the length of the Simulator.
c) The distance between the bearing locations shall be within 5% of the engine dimensions.
d) Polar Moment of Inertia (Ip) – no tolerance (required for calculation purposes only)
e) Diametral Moment of Inertia (Id) - no tolerance (required for calculation purposes only)
f) Ip-Id to be within 10% of the engine part.

7.5.1.2 General features:

a) The Simulator shall have suitable correction features so that it can be balanced about its CG.
b) Balancing correction features should be temporary and consist of threaded holes, suitable for standard bolts and weights. Axial threaded holes are preferred to radial holes as any attaching weights are less susceptible to unbalance changes with radial position. Threaded holes should be equally spaced with a minimum quantity of 12.
c) A suitable drive feature is required for either shaft driving or belt driving or both.
d) Where a toothed joint is to be included into the design, the joint shall have sufficient stiffness such that it does not distort during use. Designs with a toothed joint machined into a ring should be avoided. Where this is not possible the ring shall be securely fixed such that movement or distortion does not significantly affect the repeatability performance of the part being balanced.
e) Any moving parts on the Simulator shall be designed such that the repeatability performance at the interface is maintained.
f) Spigot joints within the Simulator design should be designed to prevent movement during use. An interference of 0.025 mm to 0.05 mm on diameter is recommended for most applications. The profile of the Simulator rotor should be designed such that there is sufficient access to the bearing positions, which may be achieved by adjusting the Ip & Id properties. Where this is not possible, the design should include a custom made roller bridge to replace the machine rollers.
g) Provision shall be made for lifting and installation of the Simulator about the CG.
h) Surfaces that interface with engine parts shall have a minimum hardness of 45 HRC to ensure satisfactory service life.
i) Bearing location surfaces that are used to locate the rotor on the balancing machine shall meet the following requirements:
   Surface finish – <0.4 microns Ra
   Surface hardness – >45 HRC
   Total run-out across the whole surface – <12 microns
j) All designs of Simulator shall include a suitable container to prevent damage when not in use. Simulator containers shall include support and restraining features suitable for all types of transportation.

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7.6 Approval of new tooling designs:

7.6.1 All new designs of tooling shall be bought-off by a suitable qualified balancing engineer. This requirement also applies to designs prepared by specialist sub-contractors.

8 CORRECTING UNBALANCE:

8.1 Correcting unbalance is achieved by: adding material, removing material or by selection of parts (e.g. blades). Components are invariably permanently corrected by removing material whereas correction of assembled rotors or modules is carried out by adding weight. Bladed discs are usually corrected by re-distribution of the blades.

8.2 When correcting unbalance on a part or assembly, all the necessary information should be specified on a drawing or solution. This should contain sufficient information to correct the part or assembly without incurring any damage. This is particularly relevant to jobs where material is removed. If a drawing or solution does not specify a method of material removal, then it will be necessary to consult a controlling laboratory. In the case of a critical part it will be necessary to obtain Critical Parts Lifing Authority approval.

Subject to process approval, the following points apply:

a) The resulting surface finish shall be equal or better than the surrounding area.
b) The method of material removal shall not overheat the surface of the job or generate a sharp feature that could lead to a fatigue mechanism.
c) When removing material from a painted surface, the surface will require restoring as per the drawing requirements. Note that there may be some health & safety issues with some paint systems and the method used to remove them.
d) Removing material whilst the part is still installed in the balancing machine is acceptable providing there is adequate provision to prevent swarf contamination of the job and surrounding machinery. Specific care shall be observed when machining parts with bearing tracks and delicate oil galleries.

8.3 Where standard weights are to be attached by the same bolts used to hold flanges together, a washer will be provided so that it can be replaced with a heavier weight of the same thickness. This avoids using different length bolts during balancing that incurs a risk of inadvertent mixing, which can lead to a reduced clamp load.

9 BLADE DISTRIBUTION METHODS.

9.1 The preferred method of distribution uses proprietary software, which sorts the input data (mass weigh or moment weight) into a pattern followed by an optimisation process for minimum unbalance. When software is not available a light/heavy sort pattern can be carried out manually, which is effective at even distribution.

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9.1.1 Light/heavy sort distribution – method:

a) Weigh or moment-weigh the blades and place in a line of ascending order from left to right.
b) If there is an odd quantity of blades remove the middle one and install it in the last remaining slot after completing the procedure.
c) If the blades are required to compensate for disc unbalance, select pairs of blades that will provide the desired quantity of unbalance and fit at the appropriate angle.
d) The blades are selected starting with the lightest followed by the heaviest, which is fitted alongside. The next lightest blade is fitted opposite the lightest blade alongside the next heaviest, which is opposite the heaviest blade. The process is repeated until all the blades are used. If required, fit the remaining odd blade. Note that this process is equivalent to Schenck Bladis NET distribution pattern 3 (2 segment heavy/light).

10 BALANCING BLADE SETS:

10.1 With some engine designs there can be a requirement to balance a blade set in isolation. This means that a blade set can be fitted in any angular orientation and if necessary changed for another set. Balancing is achieved by re-positioning the blades around a disc or dummy disc to achieve an acceptable level of unbalance. This is normally achieved using a vertical balancing machine or a moment weighing machine with the appropriate software. Note that when interchanging opposing blades, the unbalance change will be twice the unbalance difference between the two blades.

10.2 Balancing a whole blade set in a vertical balancing machine:

Where blades have no interlocking features, it should be possible to fit the complete set into a disc or dummy disc. The actual unbalance in the blade set is revealed by indexing the complete blade set relative the disc. The true unbalance can be achieved by calculation or by using the indexing facility in the balancing machine. The individual blades can then be moved around the disc to obtain the correct unbalance.

10.3 Balancing a split blade set in a vertical balancing machine:

Where blades have interlocking features that restrict the installation of the whole set, it may be possible to divide the set up into equal/even proportions. The partial sets can then be fitted to a disc or dummy disc and then indexed to reveal their actual unbalance. The process is repeated for each partial set and the total unbalance is then revealed by adding the vectors of the individual sets. Unbalance corrections can be made at any of the stages by redistributing the blades

An alternative technique involves adding clay to replicate the residual unbalance of each partial set. For a set of blades that can be divided into two equal sets, this is carried out as follows:

a) Layout the blades in ascending order and divide into two sets; odds and evens.
b) Fit the odd numbered blades to the odd numbered slots in the disc.
c) Measure the unbalance.
d) Index the set 180° and reveal the unbalance in the set of odd numbered blades. Re-distribute as necessary.
e) Remove the odd numbered blades from the disc. Use clay to re-create the residual unbalance of the odd numbered blade set.
f) Fit the even number blades to the even numbered slots and repeat the process. After indexing the calculated unbalance value represents the residual unbalance of both sets of blades. The even numbered blades can then be re-distributed to reduce the total unbalance to within limits.

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11 MOMENT WEIGHING OF COMPRESSOR & TURBINE BLADES:

11.1 Moment Weighing is instructed either as a mandatory drawing requirement or as an aid to balancing during the assembly process. There is a variety of proprietary equipment available for weighing engine blades and determining the centre of gravity. The traditional equipment utilises a beam mechanism which is counter-balanced with weights in order to achieve the correct calibration. There is also non-rotating static balancing machinery available that will achieve entirely satisfactory results. In both cases, a purposely designed fixture will be required to hold the blade about its location positions.

When used as an aid to balancing, the equipment should include a computer based system with suitable software for distributing the blades into balanced sets. The system should be capable of measuring the radial moment within 0.05% when removing and re-installing the blade. This value may be relaxed when weighing blades for assembly (aid to balancing) reasons only.

11.1.1 Advantages of Moment Weighing:

a) Provides more accurate weight data.
b) Provides better correlation with actual unbalance.
c) Permits a more accurate blade changing process.

Disadvantages of Moment Weighing:

a) Requires more sophisticated equipment.
b) Requires more sophisticated calibration methods.
c) Requires a robust tooling design to hold the blade in its natural position.
d) More time consuming, especially if the blades are to be permanently marked.
e) More opportunities to generate errors.

11.2 When conceiving a Moment Weighing process it is imperative that the correct size of machine is selected. The maximum capacity of the machine shall be such that the sensitivity is better than the measurement repeatability requirement of 0.05% (radial moment).

11.3 Where there is a mandatory requirement to weigh blades at manufacture it is imperative that the system is correctly calibrated using a working calibration blade. This is used to calibrate the moment weighing system so that comparative moment weight values can be acquired when weighing genuine engine blades. Working Calibration blades are derived from Master Calibration blades, which are kept solely as a back-up and to create Working Calibration blades.

11.4 Calibration blades shall represent (+/- 5%) the mass properties of the blades to be moment weighed. Where there are multiple part numbers of a similar standard, an applicability list shall be provided.

11.5 Calibration blades should be suitably marked such that they can be easily identified and not inadvertently mistaken for a real engine part. It is recommended that Calibration Blades are spray painted on one aerofoil surface with a brightly coloured axial line.

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11.5.1 The moment weight value of a master calibration weight is determined by either of the following methods:

a) By balance checking in its disc. The blade is installed in a disc and rotated in a balancing machine. The blade will require indexing around the disc so that the effects of the disc can be discriminated. To ensure that the blade does not tilt abnormally, the blade may run in both directions and a vector mean results obtained.
b) By installing in a moment weighing system that can accurately re-create the running radius of the blade.

Note: In both cases the blade unbalance values shall repeat within 0.05% when the blade is removed and re-installed about its location feature.

11.6 The location site for moment weighing equipment shall be carefully considered as it is imperative that the equipment is not subject to excessive environmental changes i.e. vibration, temperature and air flow. The supplier's installation requirements shall be followed precisely. Moment weighing systems shall be periodically calibrated in accordance with Appendix A.

11.7 When moment weighing a set of blades the calibration shall be checked using a Calibration blade at the beginning and the end of the weighing process. Where blades are not allocated in sets, the calibration shall be checked periodically at intervals not exceeding 50 blades or ½ a working shift. More frequent calibration checks shall be carried out when there is an environmental change (e.g., room temperature).

12 STANDARD PRACTICES:

12.1 Method of drive:

12.1.1 On horizontal machines, the drive is supplied either by a Cardan shaft or by a belt; each of which has advantages and disadvantages:

12.1.1.1 Cardan shaft:

Advantages:
- High torque load capacity
- High axial load capability
- No heat generation
- No bending influences
- Good stability

Disadvantages:
- Tooling/adaptor required
- Indexing/tool error management required
- Not suitable for small rotors
- Limited speed range

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12.1.1.2 Belt drive:

Advantages:
- Wider speed range possible
- Can be attached directly to the rotor without any tooling/adaptor, therefore avoiding indexing/tool error management
- Suitable for small rotors

Disadvantages:
- Low torque load capacity
- Can generate heat
- Can distort/bend the rotor
- More susceptible to influencing the unbalance readings
- More difficult machine loading/setting
- Axial control of the rotor is required
- Can damage the surface which is in contact with the belt
- Requires concentric/parallel surface line elements to position the belt

12.1.2 Cardan shaft – good practice guide.

When using a Cardan shaft, the following points apply:

a) To minimise unbalance effects, the Cardan shaft should be installed such that it is level and in line with the rotor.
b) When connecting the shaft, any clearance within the flange bolt holes should be taken up in the direction of rotation.
c) The use of telescopic drive shafts should be avoided as they can induce unbalance which is difficult to control.
d) The unbalance effects must be correctly managed during indexing.
e) The supplier's maintenance requirements shall be carried out as specified.
f) The Cardan shaft should be balanced to G6.3 to prevent drive shaft unbalance from affecting the part being balanced.

12.1.3 Belt drive – good practice guide.

When using a Belt Drive system, the following points apply:

a) The Belt Drive shall be set-up as specified by the supplier's operating manual.
b) The pulley diameters shall be selected such that they can't produce synchronous signals with the rotor.
c) The Belt tension must not distort or bend the rotor in any way that affects the unbalance measurements.
d) The Belt must not damage the surface of the rotor. If necessary, the rotor surface shall be protected using suitable adhesive tape.
e) The Belt should not create excessive heat in the rotor, which can affect the unbalance measurements.
f) End-stop rollers should be positioned on a clean/smooth surface thus avoiding synchronous signals.
g) The rotor should be set up such that there is a minimal axial load on the end-stop rollers. Care should be taken to ensure that axial loads do not influence the behaviour of the rotor in any way.
h) End-stop rollers should be attached to the moving part of the pedestal and not attached to the grounded part or the base of the machine. Where this is not possible, the end-stop roller should be positioned on the rotational axis of the rotor where it is not possible to induce a synchronous signal. There is an example of a suitable end-stop bung in the tool design section of this document.
i) Where the end-stop roller is positioned on the rotor, it is preferential to position it on the vertical centreline where the axial loads have less effect on unbalance readings.

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12.2 Managing rotating tooling errors:

12.2.1 During the balancing operation, it will be necessary to assess the effect of any rotating tooling. The standard method is by indexing the job relative to the tooling/drive. The balancing errors in each part can then be easily calculated using the spreadsheet calculators on the balancing website or by the proprietary software on the machine.

12.2.2 To manage tooling errors the following apply:

a) Belt driving minimises drive errors to the extent that its effects can be ignored. However, this assumes a sound set-up, which is not influenced by other factors. See the section on belt driving.
b) Cardan shaft drives are pre-balanced and invariably have low unbalance effects. However, they should be periodically checked and indexed to ensure that any errors do not have a significant effect on the job final unbalance limits
c) Where a Simulator is used, it is imperative (mandatory) that it is indexed against the job. The errors in the Simulator shall not contaminate the job and the unbalance in the job shall be repeatable on every assembly with the Simulator. Any unbalance changes that occur shall be within the final unbalance limits. Relaxation of these requirements can be achieved by statistical monitoring of tooling/repeatability errors.
d) When using the balancing machine's indexing function, the system invariably assumes that the phase reference is on the tool. Care is required when belt driving to ensure that the phase reference is on the correct part.

12.3 Recommended indexing techniques:

12.3.1.1 Technique 1. Measure the unbalance with both tooling and job at the zero/zero position. Index the job and calculate the tooling and job errors. Index the job again, back to the zero/zero position and calculate the tooling and job errors again. The repeatability can then be calculated from the two sets of data. The two calculated tool errors should then be averaged out for a more accurate final unbalance correction. See Table 1.

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TABLE 1 – TECHNIQUE 1 EXAMPLE/RECORD

OP	Details	α T-J	Left Plane gmm	Left Plane α	Right Plane gmm	Right Plane α	Remarks
1	Initial Run	0-0					
2	Job indexed relative to the tool	0-180					
3	Calculate the tool error	0-180					Add the first vector Op 1 to the second Op 2 and divide the resultant by two.
4	Calculate the job error	0-180					Subtract the first vector Op 1 from the second Op 2 and divide the resultant by two.
5	Job indexed relative to the tool	0-0					
6	Calculate the tool error	0-0					
7	Calculate the job error	0-0					
8	Calculate the average tool error	0-0					Add the first tool error to the second tool error and divide the resultant by two.
9	Calculate the repeatability of the tool error	0-0					Subtract the first tool error from the second tool error
10	Balance the job by targeting the average tool error	0-0					
11	Calculate the true job unbalance	0-0					Subtract the average tool error Op 8 from the targeted job error Op 10

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12.3.2 Technique 2. Measure the unbalance as in Technique 1, but carry out the repeatability check prior to the indexing. This technique is useful for large jobs where you want to reveal any problems with the joint as soon as possible. The disadvantage is that you only calculate one tool error. See Table 2.

TABLE 2 – TECHNIQUE 2 EXAMPLE/RECORD

OP	Details	α T-J	Left Plane		Right Plane		Remarks
			gmm	α	gmm	α	
1	Initial Run	0-0					Set-up data can be inserted here
2	Break and remake joint	0-0					
3	Average the two results	0-0					Vector mean of Op 1 and Op 2
4	Calculate the repeatability	0-0					Subtract the first vector Op 1 from the second Op 2
5	Job indexed relative to the tool	0-180					
6	Calculate the tool error	0-180					Add Op 3 to Op 5 and divide the resultant by two
7	Calculate the job error	0-180					Subtract Op 3 from Op 5 and divide the resultant by two
8	Balance the job by targeting the average tool error in Op 6	0-180					
9	Calculate the true job unbalance	0-180					Subtract the average tool error Op 6 from the targeted job error Op 8

12.3.3 Technique 3. Use the balancing machine's indexing facility. This method is useful as it compensates for the tool error and the job can be balanced out to zero instead of target balancing the tool error. The disadvantage is that only one index is carried out without any averaging of tool errors. Therefore, it is not suitable for jobs with variation in readings or large jobs that take significant time to index. Once corrected, the job will require another index to prove that it is still within the final unbalance limit. Note that the machine's indexing system should be reset so that the calculation occurs with the lower unbalance readings.

12.4 Balancing machine performance testing (on-the-job):

12.4.1 The function of all balancing systems shall be checked to ensure adequate performance. This is particularly important when commissioning new jobs or where new tooling has been designed and is part of the rotor. The following tests (in order of severity) are designed to test the performance of the machine and are applicable to all types of unbalance:

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12.4.1.1 Test 1 – unbalance detection test:

This tests the ability of the machine to detect applied mass, which shall be carried out whenever the machine is adjusted to accommodate a new job and/or whenever the machine's performance is suspect. A Testing calculator is available on the Balancing website.

The process is as follows:

a) Run the machine and record the indicated unbalance in mass units.
b) Add a known weight at any angle to the correction plane.
c) Run the machine and record the unbalance.
d) Calculate the vector difference between the two readings.
e) The difference between the actual weight added and the indicated unbalance change should be within 10%. If not, the performance may be improved by re-calibration of the machine or by carrying out a "rotor specific" calibration.

12.4.1.2 Test 2 - unbalance reduction test:

This tests the ability of the machine to remove unbalance:

a) Apply unbalance to the rotor, which is approximately equal to 10 gmm/Kg of displacement i.e. if the rotor weighs 87 Kg, then alter the unbalance until it is approximately 870 gmm.
b) Run the machine and record the indicated unbalance in mass units.
c) Apply the precise amount of correction on the rotor as demanded by the machine and record the result.
d) The indicated unbalance should reduce by at least 90%. If not, the performance may be improved by re-calibration of the machine or by carrying out a "rotor specific" calibration.

12.4.1.3 Test 3 – unbalance response (pull/target) check:

This tests the ability of the machine to identify the angle and magnitude of applied unbalance. Use the vector addition calculator on the Balancing website.

a) Run the machine and temporarily correct the rotor as best as possible with clay. Do not use the "null" function as this tests the response using a strong signal.
b) Add a known weight at a known angle and record the indicated unbalance.
c) Calculate the difference between the applied unbalance and the indicated unbalance and check that it is within 100% +/-10%. If necessary repeat the check at different angles.

12.5 Managing variation in unbalance readings:

12.5.1 During operation of the balancing machine it is imperative that stable readings are obtained. However, there are occasions when variation occurs as a result of natural behaviour of the rotor.

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12.5.1.1 Typical causes:

a) Variation due to blade movement
b) Flexibility in the rotor
c) Imperfections in the bearing surface
d) Excessive bearing clearance allowing movement of the rotor
e) Thermal expansion of the rotor due to environmental conditions
f) Whilst the rotor itself may cause instability there are other possibilities that may require checking

12.5.1.2 Variation in unbalance can be managed by:

a) Vector average over a number of runs
b) Average of readings over a period of time
c) Selection of a narrow band filtration
d) Torque control
e) Acceleration control
f) Manipulating the loose parts into a regular position
g) Lubricating the blade roots
h) Reducing bearing clearances to a minimum
a) Whilst the rotor itself may cause instability there are other possibilities that may require checking

12.5.1.3 Check list:

a) Loose objects or fluid inside the rotor
b) Loose electrical connections to the transducers on the machine
c) Loose transducers
d) Electro/magnetic contamination of the transducer circuit/wiring
e) Unstable phase reference (belt driving)
f) Incorrect selection of pedestal roller diameter causing synchronous signals
g) Incorrect selection of drive roller (belt drive) causing synchronous signals
h) Security of the pedestals and rollers
i) Level of the rotor, particularly relevant to belt driving
j) Poor condition of rollers causing uneven witness mark on the crowned surface of the roller or rollers
k) Poor end-stop control (belt driving) as a result of surface imperfections causing a once-per-rev signal
l) Position of the end-stop rollers (belt driving) where a roller may be positioned off-centre from the vertical axis of the machine
m) Anchor position of the rollers where a roller is not secured to the moving part of the pedestal
n) Poor alignment causing variation in unbalance with speed
o) Excessive bearing clearance (equipment with its own bearings)
p) Bearing contamination (equipment with its own bearings)
q) Unsuitable lubrication
r) Poor condition and alignment of machine drive shaft
s) Unstable speed control
t) Unstable phase referencing
u) Poor installation of machine causing non-linear force measurement

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12.6 Managing blade scatter:

12.6.1 Variation in balance readings due to blade scatter is caused by the blades taking up slightly different positions on successive runs. This can be managed using the following techniques:

a) Applying a suitable lubricant (approved) to the blade roots
b) Manipulating the blades into a consistent axial position that replicates an engine running position. Prior to starting the rotor may be slowly rotated and at the same time push the blades to obtain a consistent axial position. On some blade sets that have interlocking shrouds it may be necessary to lightly tap the blade roots to obtain a consistent position.
c) Averaging the balancing results using a vector mean calculator

12.7 Averaging unbalance readings over a number of runs:

12.7.1 Rotors, which show different unbalance readings between runs, can be managed by averaging the unbalance readings using a vector mean calculator. This technique works well where the variation causes a cluster of readings. On some balancing machines there is a built-in function for this task. This is particularly relevant to bladed rotors, which exhibit movement when the rotor is brought to rest. There are two methods that can be used to expose the true unbalance correction required:

12.7.1.1 Technique 1 - stop & start running of the machine between measurements:

a) Prior to starting, rotate the rotor in the direction to be balanced to the zero degree machine position.
b) Start the machine and take a reading.
c) Repeat the process adding 45º to each start, until 8 readings have been taken. If the quality of the readings is good, it is acceptable to reduce the total number of reading to 4 (starting every 90º) or 3 (starting every 120º).
d) Examine the recorded values and check that the variation in readings is forming a group that can be averaged out.
e) Using a vector mean calculator (Ref Balancing website) calculate the vector average.

12.7.1.2 Technique 2 - uninterrupted running of machine between measurements:

a) Operate the machine and determine a low speed RPM where movement of the loose blades is audibly detectable.
b) Operate the machine and carry out a series of measurements after a dwell period at the low speed RPM. A series of trials will be required to determine the dwell period and the low speed RPM.
c) The results can then be averaged out using a vector mean calculator. This technique can save a significant amount of time.

12.8 Selecting a suitable balancing speed and measurement period:

12.8.1 Selecting a suitable balancing speed is crucial to obtaining consistent readings. This can be determined by trial to find a stable region where there is a minimal unbalance change with speed change. The machine speed should be incrementally increased in 200 RPM intervals until stable i.e., when there is a minimal unbalance change. Selecting a speed at the lower end of the stable region is desirable as there will be less environmental effects (noise etc.) and less power consumed. In the case of bladed rotors, less heat will be generated.

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12.8.2 If a stable region cannot be achieved, it is likely that the unbalance change that is occurring with changes in RPM is due to a lack of alignment of the rotor. This can cause the rotational forces to generate a signal which is interpreted as unbalance. Where the unbalance change (for a 200 RPM speed increase) is greater than the final unbalance limit the following advice applies.

a) Check the machine's rollers for correct crowning. There should be a running witness in the centre of the roller's surface.
b) Rotors balanced in their own bearings shall utilise an alignment device (e.g.., Swivel Saddle) as supplied by the manufacturer of the balancing machine, which should be checked for correct operation.
c) Where the tooling design incorporates its own bearings, check that the design does not utilise a deep groove ball bearing for axial location.
d) Where the tooling design incorporates roller bearings, check that the diametrical clearance is not excessive. See the tool design section.

12.8.3 Where variation is due to loose parts (e.g. blades etc.) it may be necessary to temporarily secure the parts with tape (approved by a controlling laboratory) or remove them during the trial.

12.8.4 Where a suitable measuring speed has been determined, it will be necessary to determine a warm-up time/measuring period. When measuring unbalance for the first time or after a prolonged period of rest, the machine should be run for approximately five minutes. This can be reduced if the readings are sufficiently stable. Stability in readings is defined as unbalance variation that is less than half the final unbalance limit. The time period in which the unbalance is measured shall be sufficient to capture stable readings. It is not acceptable to capture readings when the unbalance readings are varying.

When setting up the machine, the following points apply:

a) When setting up an automatic cycle, the measurements shall only begin after a suitable settling time period, minimum recommended time period is five seconds. The minimum recommended measuring period is 20 seconds.
b) Where applicable, the filtration system should be set to average the readings over time.
c) During the commissioning stage of a job or when there are problems, the machine shall be set to capture the readings manually.

12.9 Direction of rotation:

12.9.1 On a rotor part without any blades fitted, the direction of rotation is not important, as the unbalance will be the same in either direction. However, for procedural purposes it may be simpler to specify normal machine direction, which is the same direction as the machine protractor angles or engine direction when belt driving.

12.9.2 On bladed rotors, it is convention to run compressors in engine direction and turbines in anti-engine direction. This technique attempts to load the blades as per the engine running conditions. Note that some custom designs of drives have threaded parts, which may be sensitive to direction and may loosen during running.

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12.10 Positioning rotors in the balancing machine:

12.10.1 Positioning a rotor in a balancing machine is crucial to the resolution and general performance. The following points apply:

a) When installing a rotor in a machine with open rollers the running diameter of the rotor should be within the range of the scale on pedestals.
b) Machine drive shafts should be level, as should the rotor to be balanced.
c) The running surface of the machine rollers will require the application of light machine oil.
d) During installation of a rotor that has its own bearings, bearing swivel saddles should be locked and then unlocked for balancing.
e) Assembled rotors that have static parts (e.g., compressors with their own stator vanes) will need to be installed such that the correct rotor axial clearance is maintained. It is imperative that all the necessary parts of the machine are locked to prevent axial movement. This is particularly relevant to any jacking device within the machine drive. During running, it is good practice to leave any jacking levers in a downward position so that its own weight cannot assist movement of the headstock drive.
f) Belt driving end-stops should be positioned on the vertical centre line and positioned such that there is a smooth contact with the surface of the job.

12.11 Balancing of turbine blade Retaining Plates:

12.11.1 Turbine blade Retaining Plate sets can be categorized in to three case types:

Case 1 - where the Retaining Plates are of similar length/mass to the Locking Plates and they are distributed for minimum unbalance.

Case 2 – where the Retaining Plates are not similar in length/mass to the Locking Plates and they are distributed for minimum unbalance.

Case 3 - where the Retaining Plates are not similar in length/mass to the Locking Plates and are distributed to correct for unbalance in the parent part.

12.11.2 Case 1 – the parts are mass weighed and distributed for minimum unbalance using a computer distribution programme.

12.11.3 Case 2 – the parts are mass weighed and distributed as per case 1, but with the Locking Plates evenly distributed around the set. This is necessary because the distribution programme assumes an equal angle between each plate.

12.11.4 Case 3 – the parts are mass weighed and distributed using a suitable distribution programme that takes in to account the angular variation between each plate. Alternatively, the parts can be balanced/distributed using a non-rotating static balancing machine. A suitable fixture will be required to locate the Plates at the correct radius. This method will facilitate correction of rotor unbalance and gap setting. It is also suitable for balancing the other case types.

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12.12 Balancing of magnetic rotors:

12.12.1 Magnetic rotors must be balanced on a suitable balancing machine and correct unbalance measurement shall be confirmed through detectability testing prior to acceptance of the balanced rotor.

12.12.2 A magnetic rotor can affect the measured unbalance from a balancing machine in 2 ways:

a) The magnetic field can interfere with the normal operation of the machine such as generating spurious signals due to unshielded cables.
b) The magnetic rotor can generate a spurious unbalance force through unbalanced magnetic pull (UMP) which the balancing machine interprets as mechanical unbalance.

12.12.3 There are several approaches to detecting and minimising the effects of a magnetic rotor on the balancing process.

a) Specialist non-magnetic balancing machines with shielded pickups and cables should be used where appropriate.
b) Mechanical UMP is detectable with speed checks, with changes in balancing speed causing changes in the measured effect of the UMP. Balancing at higher speeds than usual can reduce these effects on standard balancing machines.
c) Standard horizontal axis balancing machines may be less prone to the mechanical effects of UMP than standard vertical axis machines as the main UMP will be vertical downwards towards the rails of the bed and the pedestals are designed to be rigid in this direction. The effects can be reduced by using taller pedestals to increase the distance between the magnetic rotor and the rails.
d) Distance between the magnetic rotor and belt drive motor should be maximised to reduce potential interaction resulting in a horizontal pull on the rotor being balanced.

12.13 Balancing records:

12.13.1 Keeping accurate balancing records is essential to claiming compliance with the drawing/design requirements. This is particularly important when the procedures are subject to an audit or when an investigation is being carried.

12.13.1.1 The minimum information (mandatory) to be recorded is:

a) Date
b) Operator
c) Machine type No and serial No.
d) Serial No of the job (part/module).
e) Serial No of any rotating tooling (assuming there is more than one set available) used.
f) Balancing speed.
g) Direction of rotation.
h) Set-up data.
i) Initial unbalance
j) The amount of correction applied to the job.
k) Final unbalance.
l) Pre & post indexing results of any rotating tooling e.g., simulators.
m) Calculated tool errors of any rotating tooling e.g., simulators.
n) Repeatability of any interface joints.

Note: when recording unbalance angles, a suitable job datum (where applicable) should be identified so that the machine angles can be transposed (when required) to the job.

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12.14 Assembling of joints:

12.14.1 Where there is a necessity to join two parts of a rotor together at an engine interface, the relevant drawing instructions shall apply. These may refer to an appropriate Engineering Specification which contains the torque tightening details etc. Note that when carrying out the procedures it is essential that the two parts of the joint are only influenced by the torque tightening procedure. It is not acceptable to rest or lean on the parts during torque tightening. Furthermore, it may be necessary to remove any static tooling that is used for holding stators etc., so that there is no influence on the joint alignment in any way.

12.14.2 It is acceptable to reduce the bolt torque load through a joint providing that the reduced load results in comparable unbalance to the fully torqued loaded condition. Note that this comment is specifically relevant to joints that are assembled to super-torque values and where bolts are only used once per assembly.

12.14.2.1 Care shall be taken during joint assembly to ensure that any thread locking features are not damaged beyond useable limits. If necessary, shorter bolts shall be employed to avoid contact with any deformed locking threads.

12.14.3 Where a toothed joint has a known sensitivity to repeatability, it may be beneficial to pre-load the joint with a single central load, prior to applying the normal bolt loads. Invariably, this improves the initial engagement of the teeth and the overall repeatability of the joint. Applied loads should be kept to a minimum and it should not be greater than 10% of the total bolt load. The load must be released prior to applying the final bolt torque load. All pre-loading procedures will require approval by the appropriate engineering authority.

12.14.4 Where a part to be balanced interfaces with tooling that has a clearance fit, it will be necessary to take up the clearance in the same angular direction (relative the tool) every time the joint is made. Where this process fails to achieve satisfactory results, it may be possible to position the part and set true using a Dial Test Indicator (DTI); although this will have to be carried out with extreme care in order to achieve sufficient accuracy.

12.15 Repeatability of joints:

12.15.1 A typical rotor system in an engine is designed such that it can be dismantled in order to install the combustion system etc. It is therefore essential that the rotor joints at the module breakpoints give precise location. During the balancing stage it is essential that any joints are separated and re-made so that the repeatability can be verified. The unbalance change that occurs should be lower than the final unbalance limit of the job. Tooling shall be designed such that the unbalance change is minimised. When assigning a limit to a process, a nominal static unbalance value of 1 gmm/Kg (per joint) can be used. Alternatively, it is acceptable to monitor the unbalance data using SPC and set a 3 sigma Upper Control Limit.

12.15.2 Testing for repeatability:

12.15.2.1 Test 1 – checking against a limit on a drawing.

Where a limit is specified on a drawing, run the machine and record the unbalance. Separate the job from the tooling and remake the joint. Measure/record the unbalance again. Calculate the unbalance change by subtracting the first vector from the second; this value can be compared with the drawing limit. This method can be used for static and dynamic unbalance. Note that static unbalance can be calculated by vectorially adding the two dynamic readings.

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Test 2 – checking for repeatability where there is no limit quoted.

12.15.2.1.1 Sometimes there is no limit quoted but there is a need to check the repeatability to commission a process. This can be done using the same process in Test 1. The unbalance change should be approximately half the final unbalance limit of the job to be balanced.

12.15.2.2 Test 3 – defining the static repeatability in displacement (microns).

12.15.2.2.1 This is a useful method for measuring the performance of a joint during commissioning and/or where there is a significant tooling mass. First, weigh all the parts and determine the gross mass (Kg). Determine the static unbalance change as per Test 1.

12.15.2.2.2 Calculate: Displacement (microns) = Static unbalance (gmm) divided by the mass weight (Kg).

Notes:

1. Where a portion of the rotor is considered to be part of the machine drive (e.g., overhung rotor or vertical balance) its mass may be omitted from the calculation.

2. The repeatability performance of a joint is dependent on numerous factors: material properties, stiffness and joint design etc. A good performing toothed joint should repeat within one micron.

12.15.3 Check List for causes of poor joint repeatability:

a) Cleanliness. It is essential that parts are free of debris.
b) Lubrication. Most joints are assembled dry and will require degreasing.
c) Damage. Joint surfaces shall be free of defects and any visible signs of bruising and burrs etc.
d) Torque tightening. The tightening load and sequence across a joint is fundamental to repeatability. The joint should be tightened and treated the same as it is in the engine.
e) Temperature. Joints with an interference spigot joint may not be seating correctly at ambient temperature and may require the application of heat.
f) Clearance. Joints with clearance may not be positioning correctly and it may be necessary to take up the clearance in a specific direction. Some parts may benefit from clocking true at a suitable position.
g) High static load across the joint as a result of making the joint horizontally.
h) Loose parts. Variation in readings may not be caused by the joint but could be caused by a part moving during the handling process.
i) Tool wear. Tools that have worn contact surfaces can cause inconsistent seating of the job.
j) Tool design. The tool interface may have structural weakness. This is relevant to Simulators where the joint does not fully replicate the engine parts. Extra care is required with tools that have separate Curvic rings, which can distort and move.
k) Alignment. The joint shall assemble correctly without uneven external influence. This is particularly relevant with joints that are difficult to access.

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13 SIMULATORS – MAINTENANCE AND VERIFICATION REQUIREMENTS:

13.1 Prior to use, simulators must be visually examined for signs of damage or deterioration. Any defect that may affect the balancing results must be investigated and the necessary corrective action carried out.

13.2 Simulators shall be inspected and assessed against its manufacturing drawing at intervals not exceeding 12 months or after any repairs or modification. Simulators that are in frequent use shall have a reduced inspection interval, at the discretion of the local engineering authority. Simulators shall be inspected for the following features:

a) Residual unbalance. Simulators that are subject to tool error management will only require correcting when the tooling influence on the job exceeds ten times the final limit on the job or when the SPC data indicates a step change. In all other cases they should be checked and corrected in accordance with the balancing requirements on the manufacturing drawing.
b) General appearance. Simulators shall be thoroughly checked for: corrosion, deterioration of surface coatings and internal contamination with fluids or debris.
c) Bearing location surfaces. The bearing surface shall not have any defects that are visible with X5 magnification and/or can be felt with a 1mm stylus. For cases where the contact is directly on the machine's rollers, it is acceptable to repair/re-grind the bearing surface and accept any minor changes in bearing diameter. Note that changes in the bearing diameter might affect any tip grinding operations.
d) Bearing location diameters. The geometric features of the surface shall be checked to the drawing requirements. Note that the diameter is only relevant when it is part of a bearing system and dependent on correct bearing element clearance or when it is essential to a tip grinding operation.
e) Spigot interface (interference fit) diameters. The spigot diameters shall be measured for: diameter, ovality and surface finish. Wear of the diameter is acceptable providing an interference fit with its mating part is maintained.
f) Spigot interface axial location faces. Damage to the surface is acceptable providing there are no raised features that can affect the seating of the joint.
g) Toothed joints. The mating surface of each tooth shall be inspected for: bruising, dents, scratches and any defect that can affect the performance of the joint.
h) Drive location features. Spigot diameters and threaded holes shall be checked to the drawing requirements.
i) Moving parts. Functional parts (e.g., diaphragms) shall be visually examined for signs of wear and contamination. The supplier of the parts shall advise on any specific maintenance/inspection requirements and how frequently they should be performed.
j) Storage containers. Storage containers shall be checked to ensure that they are fit for purpose i.e., transportation, workshop use and long-term storage.
k) Bolted parts. All nuts and bolts shall be checked for security and any drawing torque tightening requirements observed.
l) Full disassembly and inspection of simulator components to the original manufacturing drawings is not required for maintenance. Periodic inspections should focus on the functional surfaces of the simulator that interface with parts being balanced and the balance machine.

13.3 Simulators that have passed the above checks shall be issued with appropriate documentation to demonstrate compliance with this standard. A copy of the documents shall be stored in the storage container. Attaching labels or stickers to the Simulator is not acceptable unless it can be demonstrated that the effect on residual unbalance is insignificant. Simulators that fail any of the above checks shall be corrected prior to use.

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14 PROCEDURES AND PROCESS CONTROL:

14.1 All balancing data (gmm and angle) should utilise a proforma that records all the relevant steps. This should be confined to one sheet and not segregated. A copy of the data shall be kept on the balancing section.

14.2 The local procedures for balancing tasks should contain conditional information and should avoid detailed balancing processes which are covered by a proforma.

14.3 Proforma should be composed as per the tables in section 12.3. Electronic files shall contain all the relevant SPC data, and they must be backed up on a network drive.

15 BALANCING COMPRESSOR ROTORS FOR TIP GRINDING:

15.1 Where a compressor rotor is to be balanced prior to a tip grinding operation, the complete rotor (including any rotating tooling) shall be balanced to ISO tolerance grade G2. The calculated limit of unbalance should be equally divided between two planes. Note that there is a calculator available on the balancing website.

15.2 Unbalance (gmm) = $\frac{G\,W\,30000}{\pi\,N}$

Where:

G = tolerance grade (2)

W = Total weight of the rotor (KG)

N = Tip grinding speed (rpm)

16 TRIM-BALANCING:

16.1 In principle, balancing may be carried out on any rotor providing it can be rotated and a meaningful vibration measurement observed. To carry this out satisfactorily, the following points apply:

a) Can the rotor be rotated at sufficient speed?
b) Is there a once-per-rev signal available?
c) Is the quality of the vibration signal sufficient?
d) Does the rotor respond when a test weight is applied?
e) Is there provision on the rotor to adjust the unbalance?

Note: The principles of Trim- Balancing are explained within the training material on the website. There is also a spread-sheet calculator available in the Toolbox section.

When applying a trim-balancing process on an engine, engineering approval will be required to agree the measuring process and the method of correction.

16.2 When selecting a suitable speed to run an engine, the following points apply:

a) It will be necessary to establish the effect of adding a test weight and identify any resonant conditions.
b) It may not be possible to run the engine at the optimum speeds due to vibration level limitations.
c) With a flexible rotor, it may be necessary to balance at a lower speed first, before increasing RPM where flexibility occurs.

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16.3 Self-adhesive type weights can be used to aid the calibration process - providing:

a) The risks of the weight detaching are acceptable.
b) The weight is positioned on a suitable internal surface thus preventing any radial movement.
c) The adhesive is compatible with the service environment.
d) At completion, all weights are removed and replace with the correctly configured engine parts.

Note: A controlling laboratory should be consulted to ensure that any surface preparation/cleaning is suitable/compatible with the adhesive.

16.4 Adhesive weights can be used for prolonged running on development builds – providing:

a) The risks of the weight detaching are acceptable.
b) The weight is positioned on the inside of the rotor.
c) The adhesive is compatible with the service environment.
d) The weight is subject to frequent inspections or replacement.

17 STATISTICAL PROCESS CONTROL (SPC):

17.1 SPC is an established mathematical method of analysing data, which is designed as an analytical tool to demonstrate compliance/control of a process. This can be carried out on any part/rotor being balanced but it is considered mandatory for the following:

a) Any part or rotor that is balanced with a Simulator.
b) Any part or rotor that is balanced about an engine joint interface.
c) Parts that are balanced at high speed.
d) Any part that is considered to have flexible rotor behaviour.
e) Main turbine shafts.
f) Stub Shafts that are designed for driving/supporting main rotors.
g) Any part that is considered to be sensitive to rotor dynamic behaviour.

17.2 Where a part or rotor is subject to SPC, the following data (minimum) is required to be monitored in addition to the records listed in section 12.13.1.1:

a) Initial and final unbalance.
b) Tool error management data.
c) Run-out measurements (if applicable).
d) Deflection data (high speed/flexible rotors only).

17.3 Where there are tooling errors within the rotor, a Tool Error Management plan is required for all balancing procedures that are subject to SPC. The plan shall consider all the data that is required to prove that the process is sound. When designing a management plan, the following points apply:

a) Monitoring tool errors in isolation without angular orientation shall be avoided unless it is proven not to be relevant.
b) When monitoring multiple errors, a suitable control chart shall plot the vector difference between the two values.
c) It is acceptable to waive additional indexing operations when it is seen that the calculated tool errors are in control.
d) Tool Error Management Plans shall be approved by the Rolls-Royce Technical Authority.

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17.4 All SPC monitoring shall include Shewhart Control Charts (Ref. ISO 7870-2), which display the statistical mean and upper control limit (UCL) for a data set (See Figure 3). Any data points outside the UCL should be excluded from the statistical mean. The data points shall represent a meaningful chronological period. New charts should be plotted when there are more than 5 data points and the chart should plot running data over the last 25 points. Polar charts should be used where data cannot be readily displayed in one axis (See Figure 4).

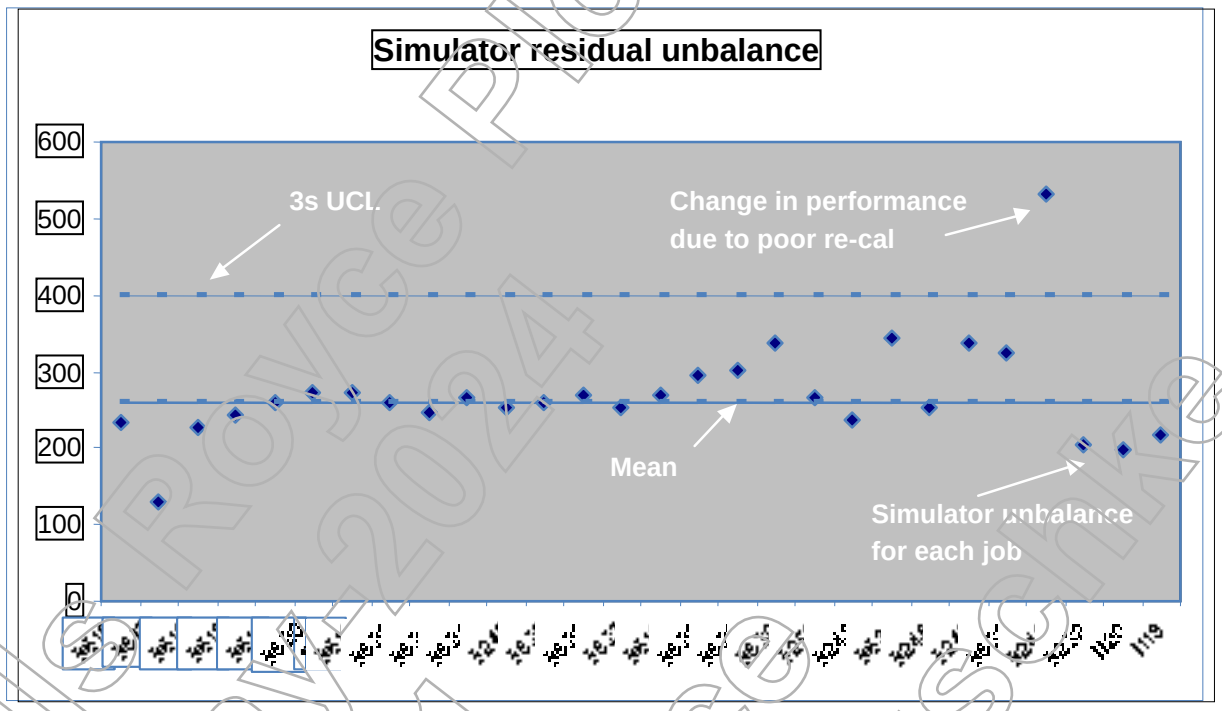

FIGURE 3 - EXAMPLE OF SHEWHART CONTROL CHART

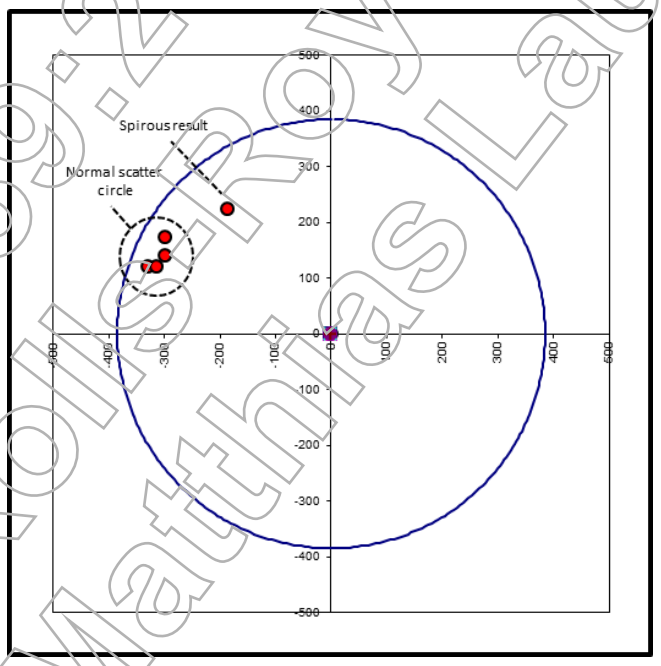

FIGURE 4 – EXAMPLE OF A POLAR CHART

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18 AUDIT/COMMISIONING REQUIREMENTS:

18.1 During the commissioning phase of a new process it is imperative that it is tested for robustness. The testing shall prove that the process is repeatable and performing correctly. This requirement is particularly important where there is new tooling involved.

18.2 The following tests apply:

a) Correct speed selection – Para. 12.8
b) Balancing machine performance – Para. 12.4
c) Repeatability of unbalance measurements between runs.
d) Repeatability of joints – Para. 12.12

18.2.1 In addition, the process should be checked for:

a) Correct machine selection i.e. machine type and size etc.
b) Periodic testing by a recognised calibration authority.
c) Adequate process instructions.
d) Adequate record keeping.

18.3 Balance process approval is granted on a part by part basis. Each part shall have a data card documenting the part specific setup and balance process.

18.4 Appendix B contains the balancing audit checklist.

19 TRAINING/SKILL REQUIREMENTS:

19.1 In order to comply with this specification it is necessary that all personnel involved in the balancing process are trained to a sufficient level. Whilst all training should be supported by an authorised training course, it is acceptable to transfer skills from a practicing engineer. The necessary skill requirements are specified in Table 3. The role requirement for final authority is specified in Table 4.

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TABLE 3 – SKILL REQUIREMENTS

Required Skills	Roles				
	Operator	Master balancer	Manufacturing specialist	Product designer	Engineering specialist
Understanding vectors	✓	✓	✓	✓	✓
Basic machine operation	✓	✓	✓		✓
Advance machine operation		✓	✓		✓
Tool error management	✓	✓	✓	✓	✓
Record keeping	✓	✓	✓		✓
Machine verification		✓			✓
Machine performance testing	✓	✓	✓		✓
Checking for repeatability	✓	✓	✓	✓	✓
Rotor specific calibration	✓	✓			✓
Job commissioning		✓	✓		✓
Speed selection		✓	✓		✓
Tool design		✓	✓		✓
Direction of rotation	✓	✓	✓		✓
Investigating problems		✓	✓		✓
Moment weighing principles	✓	✓	✓	✓	✓
Moment weighing practicals	✓	✓	✓		✓
Trim balancing					✓
Balancing specs	✓	✓	✓	✓	✓
Using .xls spreadsheets	✓	✓	✓		✓
BladisNet operation	✓	✓	✓		✓
SPC	✓	✓	✓		✓
Strategy design principles				✓	✓
Unbalance limits				✓	✓
Specifying balancing machinery			✓		✓
Flexible rotors - basic	✓	✓	✓	✓	✓
Flexible rotors - advanced		✓			✓
Training - basic workshop practice		✓	✓		✓
Training - all levels					✓

TABLE 4 – FINAL AUTHORITY

Authority	Roles				
	Operator	Master balancer	Manufacturing specialist	Product designer	Engineering specialist
Determining Balance Strategy					✓
Determining Unbalance Limits					✓
Tool drawings – buy off			✓		✓
Design definitions – buy off					✓

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20 NOTES:

20.1 References:

Shewhart Control Charts – ISO 7870-2

20.2 Revision bars indicate changes from the previous issue of this specification and are located in the left margin.

Issue	Change (ESC) Number	Change Summary (In Descending Order)	Date
B	RFS 1768	- Correction of typos - Updated balancing website address - Update of SAE Verification Test requirements - Clarification of tooling and simulator surface hardness requirement - Clarification of tooling and Cardan drive balance tolerances - Added guidance for magnetic rotors - Clarified simulator inspection - Clarified that approval is on a part by part basis - Added audit requirements	13 Oct 2023
A	ESC500034779420	- Updated scope - Revised Simulator spec - Revised calibration requirements - Revised sect on dealing with variation - New section on drive systems - New section on non-rotating machinery - New section on Simulator inspection requirements - Revised Procuring new machinery section - New section on assembling of joints - New section on balancing retaining plates - New section on Trim Balancing - New section on Tip Grinding requirements - RFS 993 and 1206	14 Feb 2019
Initial	–		19 June 12

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APPENDIX A – CALIBRATION REQUIREMENTS FOR MOMENT WEIGHING SYSTEMS

A1 INTRODUCTION:

A1.1 This document details the calibration requirements for moment weighing machinery and associated calibration weights/blades.

A2 DEFINITIONS:

Calibration weight:
A weight used during the moment weighing setting up procedure.

Master Calibration blade:
A blade which is used to certify working standard blades.

Moment weighing machine:
A device which measures the weight of an object with respect to its Centre of Gravity. Whilst it is common to measure in one direction (radial), large fan blades can be measured in three directions (radial, tangential and axial). The standard SI unit of measurement is gmm although other units may be used in practice.

Working Calibration blade:
A blade which has been calibrated against a Master Calibration blade and is used for calibrating the moment weighing system, as required.

A3 STORAGE REQUIREMENTS:

A3.1 All calibration blade standards shall be stored in a suitable container such that they cannot deteriorate in any way, which affects the moment weight value. In addition, the blades shall be stored in a normal workshop environment and not subject to adverse conditions. Storage containers shall be suitably marked with description of the calibration blade.

A4 APPLICABILITY:

A4.1 Calibration weights and blades shall be checked at intervals not exceeding:

- Calibration weights – 5 years
- Master calibration blades – 12 months
- Working calibration blades – 12 months

A4.2 Moment weighing machines shall be checked when:

- The machine is first installed and commissioned.
- Any repairs that are carried out that could affect the calibration.
- Whenever the machine is relocated.
- At intervals not exceeding 12 months.

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A5 MOMENT WEIGHING MACHINES:

A5.1 Moment weighing machines shall be checked for calibration in accordance with the manufacturer's requirements. This is detailed in the manufacturers support documentation and requires the machine to be tested throughout its range of measurement.

A5.2 The acceptance criteria for the test results vary with the machine's manufacturer and size etc. This information is detailed in the specification section of the manufacture's manual.

A6 EVALUATION AND TESTING OF CALIBRATION WEIGHTS:

A6.1 The calibration weights used for testing moment weighing machines shall be checked for the following:

- Visual deterioration. Any defect that could lead to a change in mass properties is unacceptable and shall be corrected.
- Mass. Each weight shall have its mass verified and recorded by a suitable metrology authority. The accuracy of the measurement is dependent on the resolution of the moment weighing system, as advised by the manufacturer of the machine.

A7 EVALUATION AND TESTING OF CALIBRATION BLADES:

A7.1 All types of calibration blade shall be visually examined for the following defects:

- Deterioration that could lead to a change in mass moment weight.
- Wear/damage of the blade location feature.
- Deterioration of the blade marking that could lead to the blade being inadvertently mistaken for a real engine part. See section 11.5.

A7.2 The mass weight shall be checked to ensure that it has not changed by more than 0.05% in between calibration intervals.

A7.3 The mass moment weight of Master and Working blades shall be checked relative to one another. The mass moment weight values shall be within 0.05%.

A8 CALIBRATION RECORDS:

A8.1 Calibration records shall be kept throughout the working life of the machine. At a minimum, the following information shall be recorded:

- The manufacturer/supplier of the machine.
- Machine type and serial number.
- Calibration blade part/serial numbers.
- Calibration weight part/serial numbers.
- Any relevant set-up data for the machine.
- All calibration results and date of the testing.
- Any adjustments carried out.
- Name of the person who performed the calibration.
- Any other deviations or relevant information.

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APPENDIX B – AUDIT CHECKLIST

Supplier:
Date:
Auditor:

Section	Question	Yes/No/Na	Comment
6	**Balancing Machine – General Practices**		
6.1	**Balancing Machine selection**		

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6.1.1	The following points apply during the planning stage of the balancing process: a) What type of balancing machine? Hard or soft? c) Does balancing process work within specification range of the machine? d) Is the bearing journal diameters of the job within the range of the scale on the machine's Roller Bridge? e) Where a job is to be shaft-driven, is the rotor assembly inline and level with the machine drive? Any effect on the unbalance measurements due to the small inclination? f) Does the measuring system have the sufficient functionality for the planned job e.g. indexing & averaging etc.? g) Has the machine (hard bearing type) been calibrated? h) Does the balancing machine drive system control RPM and acceleration adequately?		
6.2	**Balancing machine installation**		
6.2.1	Does the balancing machine (hard type) attach to a sufficiently sized mass/inertia block?		
6.2.2	Alternatively, does the balancing machine install directly to the floor slab?		
	Does the slab or base's mass and density meet the requirements of the manufacture/supplier of the machine?		

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	Does the distance from site machine to the edge of a floor slab meet the minimum dimension requirement?		
6.2.2.1	Where is the balancing machine located? Is the location affected by high levels of vibration?		
6.2.2.2	Does the balancing machine installation meet the manufacturer's requirement? Has the supplier discovered any cracks or structural weakness on the floor bed?		
6.2.2.3	Does the balancing machine conform to all the relevant health and safety requirements? Is the balancing machine positioned inside an acoustic cabinet? Does the acoustic cabinet influence the measurement of unbalance and inhibit inspection of the machine bed?		
6.3	**Balancing machine calibration requirements**		
6.3.1	Are all balancing machines (hard bearing type) calibrated with a proving rotor at intervals not exceeding 12 months?		
	What is the size and mass of the Proving Rotor? Is it representative of the average mass of the job being balanced?		
	What is the range in the job mass?		

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	Does the calibration comply with either of the following standards: a) ISO 21940-21 b) SAE AS 8617		
6.3.2	Are moment weighing machines calibrated in accordance with appendix A at intervals not exceeding 12 months?		
6.4	**Balancing machine maintenance**		
6.4.1	During day to day usage, does supplier check the following items to monitor wear and damage on machinery: a) Check the pedestal rollers for wear and uneven running? b) Check the bearing housing saddles for general damage? c) Check drive shaft joint?		
6.5	**Selecting machine rollers**		
6.5.1	What types of rollers are used for the balancing machine? Are the rollers crowned?		
6.5.2	Is the roller diameter within 10% of the bearing journal diameter on the job or exactly 50% of the diameter on the job?		
7	**Tool design**		
7.1	**General principles**		
7.1.1	Does supplier comply to the following design principles:		

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	a) Rotating tooling shall be completely rigid and have no effect on the balancing results.		
	b) Rotating tooling shall be designed such that unbalance errors can be identified and managed. The tooling shall be capable of indexing relative to the job a suitable angle (ideally 180º) so that its errors can be compensated for.		
	c) Surfaces that abut engine parts shall have a minimum hardness of 45 HRC in order to control wear within its design life.		
	d) Tooling shall be designed such that it does not damage the job to be balanced in any way.		
	e) The gross mass of rotating tooling (excluding Simulators) should not exceed the mass of the rotor. Where this is not possible, the gross mass should be kept to a minimum.		
	f) The parasitic mass of static parts should be kept to a minimum.		
	g) Painted parts shall be of sufficient quality to prevent flaking and contamination of working parts.		
	h) Plain bearings shall not be used for locating the rotor in hard-bearing balancing machines.		
	i) Deep-groove ball bearings shall not be used for locating the rotor in hard-bearing balancing machines.		

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	j) Roller type bearings can be used in tooling designs providing they do not have excessive clearance resulting in movement of the rotor. Diametrical clearance should be between 0.013mm and 0.039mm. Note that bearing assemblies designed for engine use have larger clearances and in general are not suitable for balancing equipment at lower speeds.		
	k) Self-contained bearings used in balancing equipment shall not have free location bearing elements i.e. the bearing elements shall be located in a cage.		
	l) Static and rotating tooling for hard bearing balancing machines shall not have a natural frequency within the running range of the machine; ideally the natural frequency should be at least 3 times the balancing speed.		
	m) Rotating tooling shall not be designed with moveable parts that affect the repeatability. Securing clamps etc. shall be located such that any unbalance change does not affect the ability to meet the final unbalance limit.		
	n) Bearing location surfaces that are used to locate the rotor on the balancing machine shall meet the following requirements: Surface finish – <0.4 microns Ra Surface hardness – >45 HRC Total run-out across the whole surface – <12 microns		

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	o) Any restraining feature that stores energy e.g. under tensile preload or hydraulic pressure shall be designed such that failure of any part is contained. Where this is not possible, the design shall be subject to a thorough risk analysis in order to satisfy health and safety requirements.		
	p) Rotating tooling shall be balanced such that the tooling influence on the job is less than ten times the final per plane balance limit on the job.		
	q) Rotating tooling used for dynamic balancing shall have two correction planes to correct for couple imbalance.		
7.2	**End-stop control bungs**		
7.2.1	Does supplier use bung for setting end-stop rollers of the machine?		
	If yes, what does it made from?		
	Does the bung have suitable tolerances so that it can be fitted easily in an internal or external diameter?		
7.3	**Balancing machine roller bridges – design features**		
7.3.1	Does the supplier use a roller bridge? If yes, does it comply to the following design principles:		
	a) The minimum tolerance class of the bearings shall conform to ISO 492 class 6.		

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	b) The lubricant used within the bearing should not be packed with grease that has a viscosity greater than 2 (NLGI) as a higher number is likely to cause skidding.		
	c) The contact surface of the rollers shall be crowned. The outside diameter of the bearing should have a 1500 mm crown radius. On smaller bearing diameters (less than 35mm approx.) the crown radius may be reduced to 1000mm. Where bearings have a reduced width (less than 10mm approx.) the crown radius should be reduced to 500mm to ensure that the rotor contact surface does not contact the edges of the bearing surface. The bearings shall be checked to ensure that there are no resonant frequencies within the operating range of the machine.		
	d) The bearing supports should be permanently fixed and have sufficient stiffness to prevent any measurable deflection of the job. Failure to meet this requirement is likely to corrupt the unbalance readings necessitating a rotor specific calibration of the job.		
	e) Bearing support pillars should have a heavy interference with the bearing internal diameter of 0.001 mm per 1.0 mm of diameter.		
	f) The fixing centres of the bearings should be designed such that the angular relationship with the rotor axis is equal (i.e. 60º).		
7.4	**Moment weighing fixtures for blades**		

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7.4.1	Does the supplier use a moment weigh fixture? If yes, does it comply to the following design principles:		
	a) Is the blade located such that the correct moment is measured?		
	b) Are blades with dovetail or pin type roots loaded outwards to ensure measurements are representative of the engine running condition?		
	c) Do blades with fir-tree type roots locate on opposing knife-edges or pins providing they contact a working surface of the root?		
	d) Where an absolute unbalance measurement is required, is the fixture design such that measurements are taken with the blade at its running radius?		
	e) Does the measurement repeat within 0.05% while removing and re-fitting the blade to the fixture?		
	f) Does the contact surface of the fixture have sufficient hardness to avoid wear-out and variation in readings?		
	g) Do any moving parts of the fixture have any significant effect on the measurements?		
7.5	**Simulators - requirements**		
7.5.1	Does the drawing or solution specify the required properties where a design definition requires a simulator? If yes, does it comply to the following design principles:		

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	a) Is the mass within 5% of the design definition requirement?		
	b) Is the axial centre of gravity within 5% of the length of the Simulator?		
	c) Is the distance between the bearings locations within 5% of the engine dimensions?		
	f) Is Ip-Id within 10% of engine part?		
7.5.1.2	Does simulator have the following design features:		
	a) Does the simulator have suitable correction features so that it can be balanced about its CG?		
	b) Do balancing correction features consist of threaded holes? Are the threaded holes are equally spaced with a minimum quantity of 12?		
	c) What type of drive feature is used by supplier? Shaft driving or belt driving or both?		
	d) Does the joint have sufficient stiffness such that it has no distortion during use?		
	e) Do the moving parts on the simulator maintain performance repeatability?		
	f) Do the spigot fits within the simulator design have a heavy interference of 0.025 to 0.05 mm per 1.0mm of diameter? Is the profile of the simulator rotor designed such that there is sufficient access to the bearing positions?		

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	g) Does simulator have provision for lifting and installation of the simulator about the CG?		
	h) Do Surfaces that interface with the engine parts have a minimum hardness of 45 HRC?		
	i) Bearing location surfaces that are used to locate the rotor on the balancing machine shall meet the following requirements: Surface finish – <0.4 microns Ra Surface hardness – >45 HRC Total run-out across the whole surface – <12 microns		
	j) Does the simulator include a suitable container to prevent damage when not in use? Simulator containers shall include support and restraining features suitable for all types of transportation.		
7.6.1	Have all new tooling designs been bought off by a suitable qualified balancing engineer?		
8	**Correcting unbalance**		
8.1	How does supplier correct unbalance in the part or assembly?		

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8.2	Does the drawing or work instruction contain sufficient information to correct the part or assembly without incurring any damage? If a drawing or solution does not specify a method of material removal, does supplier consult to controlling laboratory? In the case of a critical part, does supplier obtain critical parts lifting authority approval if a drawing or solution does not specify a method of material removal?		
	Subject to process approval, the following points apply: a) Is the resulting surface finish equal or better that the surrounding area? b) Does the method of material removal overheat the surface of the job or generate a sharp feature that could lead to a fatigue mechanism? c) Does the surface of the job require restoring as per the drawing requirement when supplier removes material from a painted surface? d) Does supplier remove material whilst the part is still installed in the balancing machine?		
8.3	Does supplier use washers where standard weights are to be attached by the same bolts used to hold flanges together?		
9	**Blade distribution methods**		
9.1	Does supplier use software to achieve minimum unbalance?		

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9.1.1	If software is not available and supplier is using a Light/heavy sort distribution-method: a) Weight or moment-weight the blades and place in a line of ascending order from left to right. b) If there is an odd quantity of blades remove the middle one and install it in the last remaining slot after completing the procedure. c) If the blades are required to compensate for disc unbalance, select pairs of blades that will provide the desired quantity of unbalance and fit at the appropriate angle. d) The blades are selected starting with the lightest followed by the heaviest, which is fitted alongside. The next lightest blade is fitted opposite the lightest blade alongside the next heaviest, which is opposite the heaviest blade. The process is repeated until all the blades are used. If required, fit the remaining odd blade.		
10	**Balancing Blade Sets**		
10.2	Does supplier balance a whole blade set in a vertical balancing machine where blades have no interlocking feature?		
10.3	Does supplier balance a split blade set in a vertical balancing machine where blades have interlocking features?		

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10.3 cont'd	Does supplier use alternative technique involves adding clay to replicate the residual unbalance of each partial set? For a set of blades that can be divided into two equal sets, is it carried out as follows: a) Layout the blades in ascending order and divide into two sets; odds and evens. b) Fit the odd numbered blades to the odd numbers slots in the disc. c) Measure the unbalance. d) Index the set 180° and reveal the unbalance in the set of odd numbered blades. Re-distribute as necessary. e) Remove the odd numbered blades from the disc. Use clay to re-create the residual unbalance of the odd numbers blade set. f) Fit the even number blades to the even numbered slots and repeat the process. After indexing the calculated unbalance value represents the residual unbalance of both sets of blades. The even numbered blades can then be re-distributed to reduce the total unbalance to within limits.		

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11	**Moment weighing of compressor and turbine blades**		
11.1	Is moment weighing instructed either as a mandatory drawing requirement or as an aid to balancing during assembly process? When used as an aid to balancing, does the equipment include a computer based system with suitable software for distributing the blades into balanced sets? Is the system capable of measuring the moment weight such that it is repeatable within 0.05% when removing and re-installing the blade?		
11.2	Is the maximum capacity of the machine sensitivity better that the measurement repeatability requirement of 0.05%?		
11.3	Is the system is correctly calibrated using a working calibration blade?		
11.4	Does the calibration blade represent (+/- 5%) the mass properties of the blades to be weighed?		
11.5	Is the calibration blade suitably marked?		
11.5.1	Which method is used to determine the moment weight value of a master calibration weight? a) By balancing check on disc. b) By installing in a moment weighing system that can accurately re-create the running radius of the blade.		

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11.6	Is the location for moment weighing equipment subject to excessive environmental changes i.e. vibration, temperature and air flow? Is the weighing system calibrated periodically in accordance with Appendix A?		
11.7	When moment weighing a set of blades, is the calibration is checked by using a calibration blade at the beginning and the end of the weighing process?		
	Where blades are not allocated in sets, is the calibration is checked at intervals not exceeding 50 blades or 0.5 working shift?		
12	**Standard practices:**		
12.2	**Managing rotating tooling errors**		
12.2.2	How does supplier manage tooling error? Does 12.2.2 (a) to (d) apply?		
12.3	Which technique does supplier use for indexing? Techniques 1, 2 or 3?		
12.4	**Balancing machine performance testing (on-the-job)**		
12.4.1	Is the function of all balancing system checked to ensure adequate performance? Does supplier use test 1, test 2 or test 3 requirements?		

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12.5	**Managing variation in unbalance readings:**		
12.5.1.2	How does supplier manage the variation in unbalance?		
12.5.1.3	Does supplier have a checklist to prevent variation in unbalance?		
12.6	**Managing blade scatter:**		
12.6.1	What techniques does supplier use to manage the variation in balance readings due to blade scatter?		
12.7	**Averaging unbalance readings over a number of runs:**		
12.7.1	Which method supplier does use to expose the true unbalance correction? Technique 1 or technique 2?		
12.8	**Selecting a suitable balancing speed**		
12.8.1	What is the balancing speed? Was the speed increased incrementally in 200RPM intervals until stable?		

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12.8.2	Where the unbalance change (for a 200 RPM speed increase) is greater that the final unbalance limit, does supplier apply the following item: a) Check the machine's rollers for correct crowning. There should be a running witness in the Centre of the roller's surface. b) Rotors balanced in their own bearings shall utilize an alignment device (e.g., swivel saddle) as supplied by the manufacture of the balancing machine, which should be checked for correct operation. c) Where the tooling design incorporated its own bearings, check that the design does not utilise a deep grove ball bearing for axial location. d) Where the tooling design incorporates roller bearings, check that the diametrical clearance is not excessive. See the tool design section.		
12.9	**Direction of rotation**		
12.9.1	Is the machine direction of rotation appropriate for bladed rotors?		

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12.10	**Positioning rotors in the balancing machine:**		
12.10.1	Does supplier apply following points while positioning a rotor in a balancing machine: a) When installing a rotor in a machine with open rollers the running diameter of the rotor should be within the range of the scale on pedestals. b) Machine drive shafts should be level, as should the rotor to be balanced. c) The running surface of the machine rollers will require the application of light machine oil. d) During installation of a rotor that has its own bearings, bearing swivel saddles should be locked and then unlocked for balancing. e) Assembled rotors that have static parts (e.g. compressors with their own stator vanes) will need to be installed such that the correct rotor axial clearance is maintained. It is imperative that all the necessary parts of the machine are locked to prevent axial movement. This is particularly relevant to any jacking device within the machine drive. During running, it is good practice to leave any jacking levers in a downward position so that its own weight cannot assist movement of the headstock drive. f) Belt driving end-stops should be positional on the vertical Centre line and positioned such that there is a smooth contact with the surface of the parts.		

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12.11	**Balancing of turbine blade retaining plates**		
12.11.1	Are turbine blade retaining plates included in the assembly balance? If so, case 1 case 2 or case 3 apply.		
12.12	**Balancing of magnetic rotors**		
12.12.1	Was detectability test performed on magnetic rotor?		
12.13	**Balancing records:**		
12.13.1	Does the supplier keep balancing records? Where are the records kept?		

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12.13.1.1	Do the balancing records contain the following minimum information: a) Date b) Operator c) Machine type and serial no. d) Serial no of the job (part/module). e) Serial no. of any rotating tooling (assuming there is more than one set available) used. f) Balancing speed. g) Direction of rotation. h) Set-up data. i) Initial unbalance. j) The amount of correction applied to the job. k) Final unbalance. l) Pre & post indexing results of any rotating tooling e.g. simulators. m) Calculated tool errors of any rotating tooling e.g. simulator. n) Repeatability of any interface joints.		
12.14	**Assembling of joints**		
12.14.1	Are assembly instructions required for the parts being balanced?		
12.14.4	Are any clearance fits present in the tooling that need to be taken up in a consistent angular direction?		

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12.15	**Repeatability of joints**		
12.15.2	Which techniques does supplier use for checking repeatability of joints? Are test 1, test 2, or test 3 used?		
12.15.3	Does supplier have a checklist to avoid poor joint repeatability? a) Cleanliness. b) Lubrication. c) Damage. d) Torque tightening. e) Temperature. f) Clearance. g) High static load across the joint. h) Loose parts. i) Tool wear. j) Tool design. k) Alignment.		
13	**Simulators – Maintenance and verification requirements**		

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13.2	Simulators shall be inspected at intervals not exceeding 12 months for the following features: a) Residual unbalance. b) General appearance. c) Bearing location surfaces. d) Bearing location diameters. e) Spigot interface diameters. f) Spigot interface axial location faces. g) Toothed joints. h) Drive location features. i) Moving parts. j) Storage containers. k) Bolted parts.		
13.3	Does supplier have documentation of simulator inspection in the simulator storage container?		
17	**Statistical process control (SPC)**		
17.2	Where a part or rotor is subject to SPC, does supplier monitor the following data (minimum): a) Initial and final balance. b) Tool error management data. c) Run-out measurements (if applicable) d) Deflection data (high speed/flexible rotors only).		

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17.3	Does supplier have a tool error management plan where there are tooling errors within rotor? Does the tool error management plan comply to the following points: a) Monitoring tools errors in isolation without angular orientation shall be avoided unless it is proven not to be relevant. b) When monitoring multiple errors, a suitable control chart shall plot the vector difference between the two values. c) It is acceptable to waive additional indexing operations when it is seen that the calculated tool errors are in control. d) Tool error management plans shall be approved by Rolls-Royce Technical Authority.		
17.3	Does all SPC monitoring include Shewhart control chart?		
18	**Audit/commissioning requirements:**		
18.2	During the commissioning phase of a new process, does supplier perform following testing: a) Correct speed selection. b) Balancing machine performance. c) Repeatability of unbalance measurements between runs. d) Repeatability of the joints.		

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18.2.1	Does the balancing process is checked as followings: a) Correct machine selection i.e. machine type and size etc. b) Periodic testing by a recognized calibration authority. c) Adequate process instructions. d) Adequate record keeping.		
18.3	Does supplier have separate data cards for each part being balanced?		
19	**Training/skill requirements:**		
19.1	Are supplier balancing personnel are trained in accordance with table 3 – skill requirements?		
Appendix A	**Calibration Requirements for moment weighing systems**		
A3.1	Are all calibration blade standards stored in a suitable container such that they cannot deteriorate in any way? What is the storage environment? Are the storage containers marked with a description of the calibration blade?		

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A4	**Applicability**		
A4.1	Are the calibration weights and blades checked at intervals not exceeding: • Calibration weights – 5 years • Master calibration blades – 12 months • Working calibration blades – 12 months		
A4.2	Are the moment weighing machines checked when: • The machine is first installed and commissioned. • Any repairs that are carried out that could affect the calibration. • Whenever the machine is relocated. • At intervals not exceeding 12 months.		
A5	**Moment weighing machines:**		
A5.1	Are the moment weighing machines checked for calibration in accordance with the manufacturer's requirements?		
A6	**Evaluation and testing of calibration weights:**		
A6.1	Are the calibration weighs used for testing moment weighing machine checked for the following: • Visual deterioration. • Mass.		

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A7	**Evaluation and testing of calibration blades:**		
A7.1	Are all types of calibration blade visually examined for the following defects: • Deterioration that could lead to a change in mass moment weight. • Wear/damage of the blade location feature. • Deterioration of the blade marking that could lead to the blade being inadvertently mistaken for a real engine part.		
A7.2	Was the mass weight checked to ensure that it has not changed by more than 0.05% in between calibration intervals?		
A7.3	Is the mass moment weight of master and working blades checked relative to one another? The mass moment weight values shall be within 0.05%		

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A8	**Calibration records**		
A8.1	Are the calibration records kept for the working life of the machine? Do the records contain the following minimum information: • The manufacturer/supplier of the machine. • Machine type and serial number. • Calibration blade part/serial numbers. • Calibration weight part/serial numbers. • Any relevant set-up data for the machine. • All calibration results and date of the testing. • Any adjustment carried out. • Name of the person who carried out the calibration. • Any other deviations or relevant information.		

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